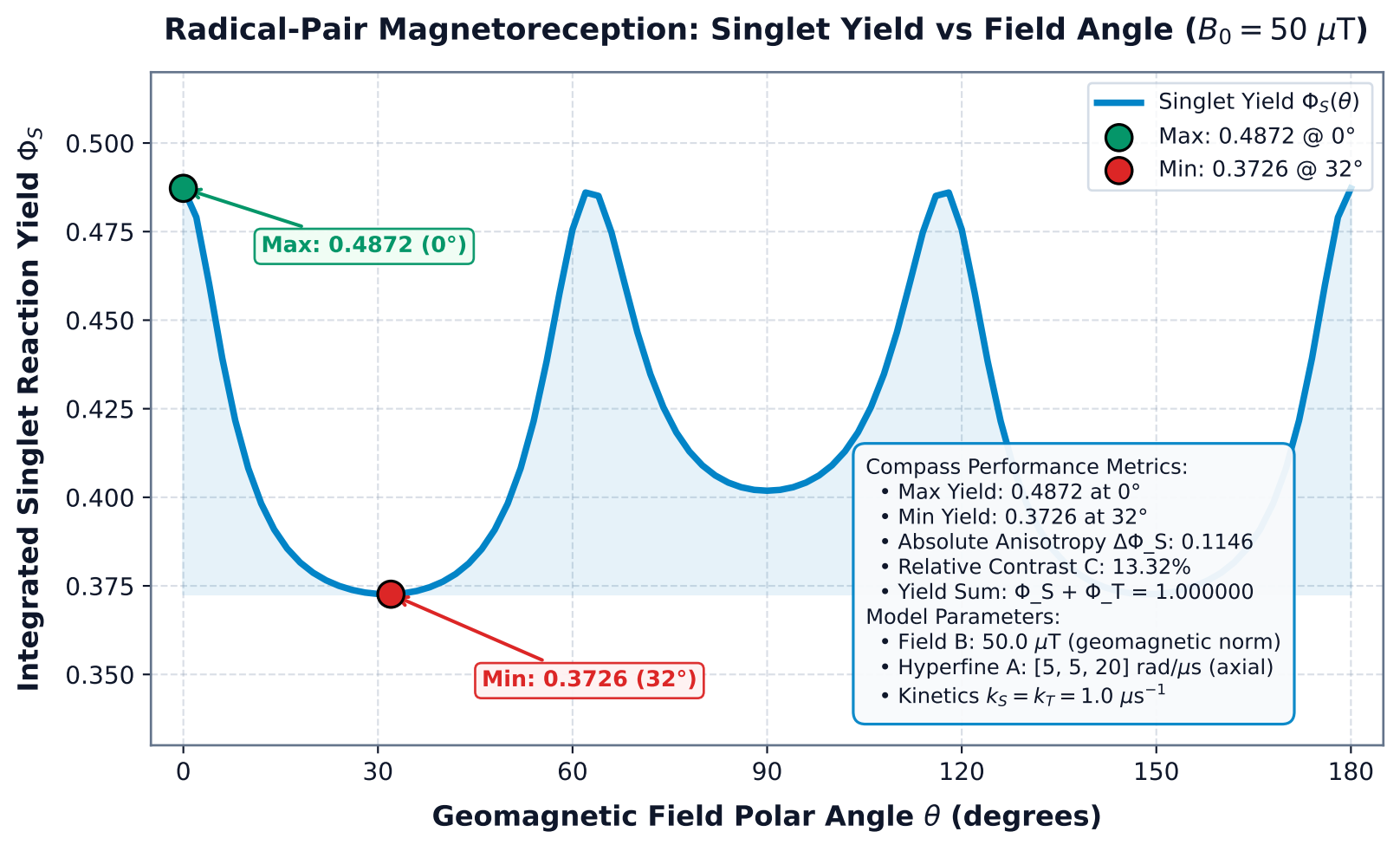
Radical-Pair Magnetoreception: Singlet Yield vs Field Angle ($B_0 = 50\ \mu$T)
Integrated Singlet Reaction Yield Φ_S
0.500
0.475
0.450
0.425
0.400
0.375
0.350
0
30
60
90
120
150
180
Geomagnetic Field Polar Angle θ (degrees)
Singlet Yield $\Phi_S(\theta)$
Max: 0.4872 @ 0°
Min: 0.3726 @ 32°
Max: 0.4872 (0°)
Min: 0.3726 (32°)
Compass Performance Metrics:
• Max Yield: 0.4872 at 0°
• Min Yield: 0.3726 at 32°
• Absolute Anisotropy ΔΦ_S: 0.1146
• Relative Contrast C: 13.32%
• Yield Sum: Φ_S + Φ_T = 1.000000
Model Parameters:
• Field B: 50.0 μT (geomagnetic norm)
• Hyperfine A: [5, 5, 20] rad/μs (axial)
• Kinetics $k_S = k_T = 1.0\ \mu s^{-1}$